HEALTHCARE ANALYTICS PROJECT

Vitalis

Healthcare data, transformed into decisions.

Six Power BI dashboard pages, a corrected SQL workbook, and a React web app — all built on a synthetic patient dataset of 20,000 appointments across 5 clinics.

PROJECT DOCUMENTATION

What this project is

Vitalis is a healthcare analytics project that analyses patient, financial, and operational data using Power BI dashboards. The objective is to understand patient demographics, service utilisation, financial performance, and operational efficiency across multiple clinic locations and medical specialties.

The project transforms raw healthcare data into meaningful insights that can support better decision-making and improved healthcare service delivery.

AT A GLANCE

$6M	Total revenue
5,000	Patients
20,000	Appointments
15,000	Diagnoses
5	Clinics

What the dashboards cover

Five comprehensive views built on the same underlying data — each one designed to answer a different question about the practice.

FINANCIAL

Money

Total revenue (~$6M), patient payments (~$3M), revenue by location, by specialty, by age group.

GEOGRAPHIC

Place

Patient distribution across Seattle, Spokane, Tacoma, Portland, and Olympia.

CLINICAL

Specialty

Appointment activity and revenue by Orthopedics, Cardiology, Oncology, Neurology, Pediatrics, General Medicine.

DEMOGRAPHIC

People

Gender, age group, insurance provider, and city-based distributions of the patient base.

OPERATIONAL

Activity

Appointments by status (Completed / Cancelled / No-show), by month, by day of week, and diagnosis counts.

Six tables, one fact table

Appointments is the central fact table. Billing, diagnoses, and medications all reference it via foreign keys.

patients

patient_id
first_name
last_name
gender · dob
city · insurance

billing

bill_id
appointment_id ↗
amount
insurance_covered
patient_paid

appointments

appointment_id
patient_id ↗
doctor_id ↗
date · status
visit_reason

doctors

doctor_id
doctor_name
specialty
clinic_location

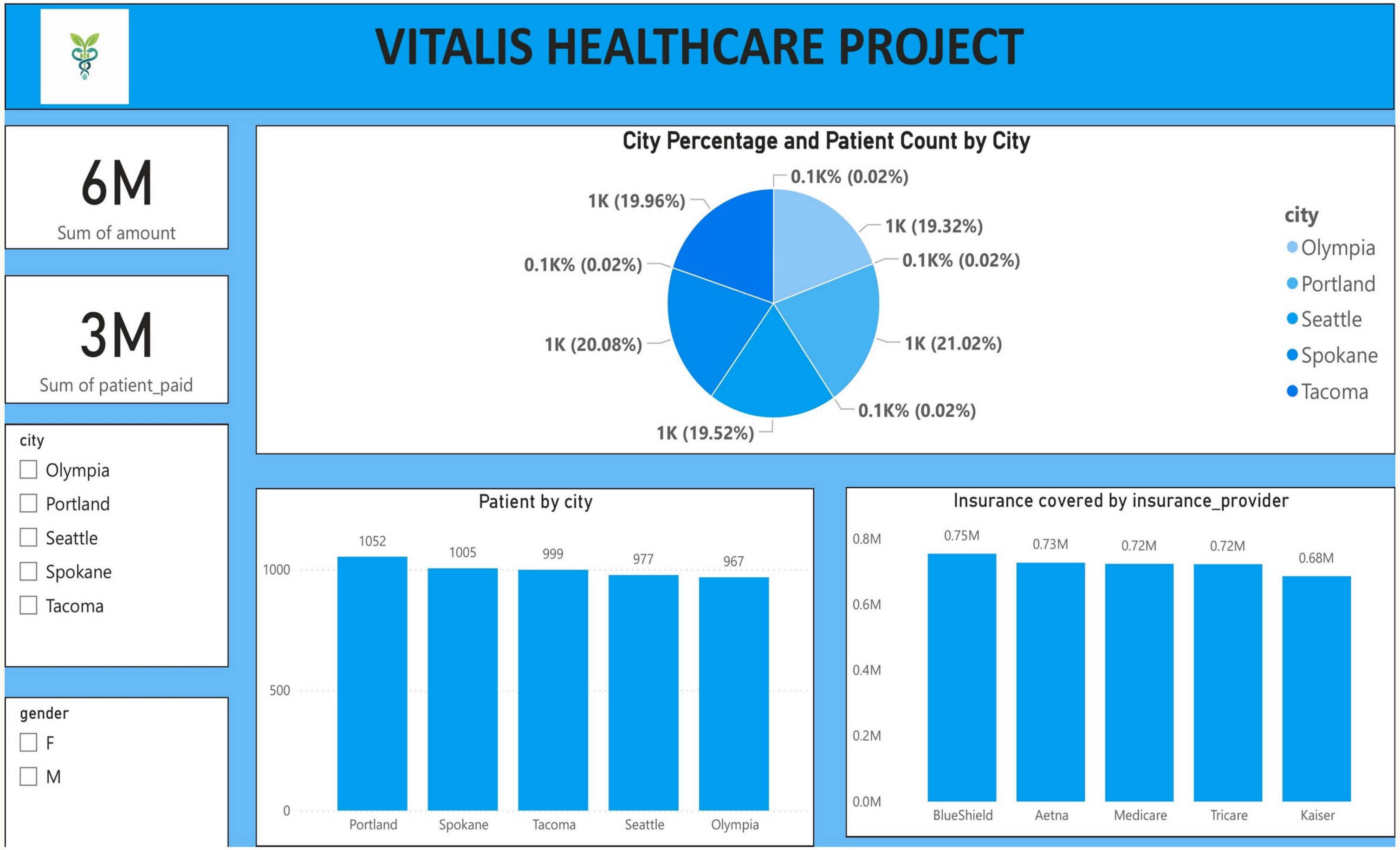
medications

med_id
patient_id ↗
medication_name
dosage
start_date · end_date

diagnoses

diagnosis_id
appointment_id ↗
diagnosis_code
diagnosis_description

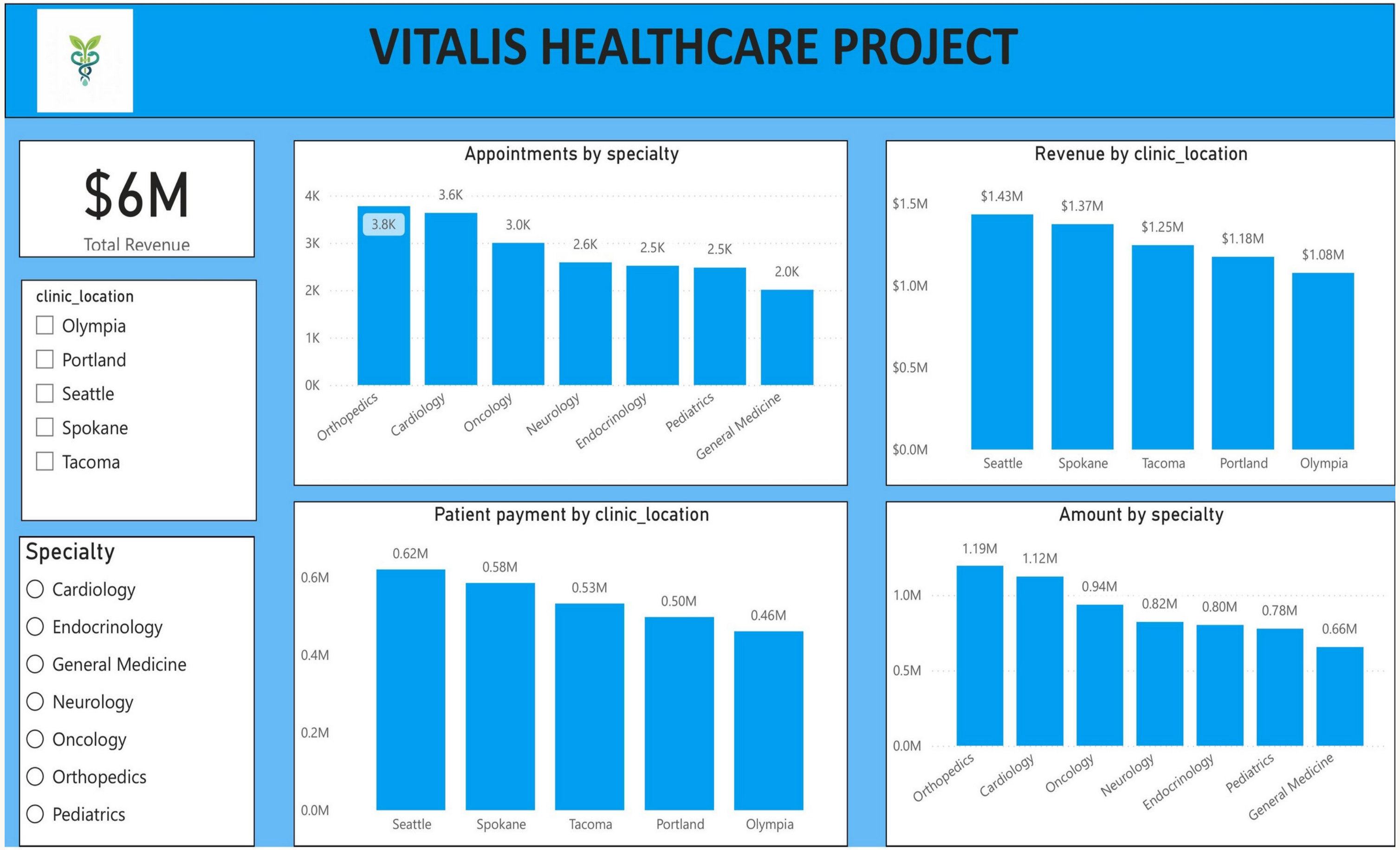
Who the patients are



KEY OBSERVATIONS

1. 5,000 patients across 5 cities — almost evenly distributed
2. Portland leads with 1,052 patients; Olympia trails at 967
3. Insurance coverage is well-spread: BlueShield ($0.75M) just edges Aetna ($0.73M)
4. Gender split is balanced (F / M) across every city

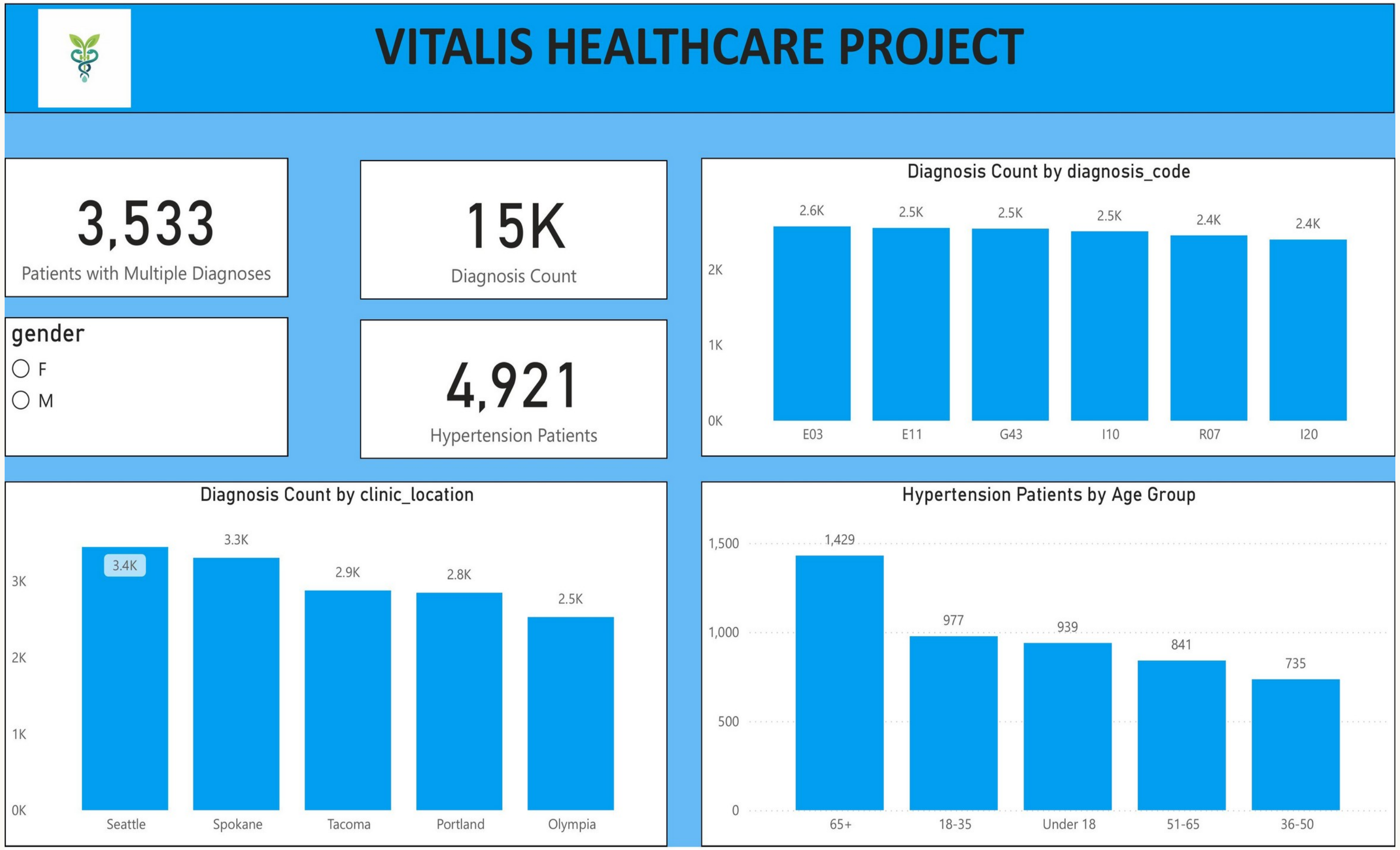
Where the revenue comes from



KEY OBSERVATIONS

1. Total revenue: $6M across all specialties and clinics
2. Orthopedics is the top specialty by revenue ($1.19M) and volume (3.8K appointments)
3. Seattle leads clinic revenue at $1.43M; Olympia trails at $1.08M
4. General Medicine is lowest-grossing at $0.66M

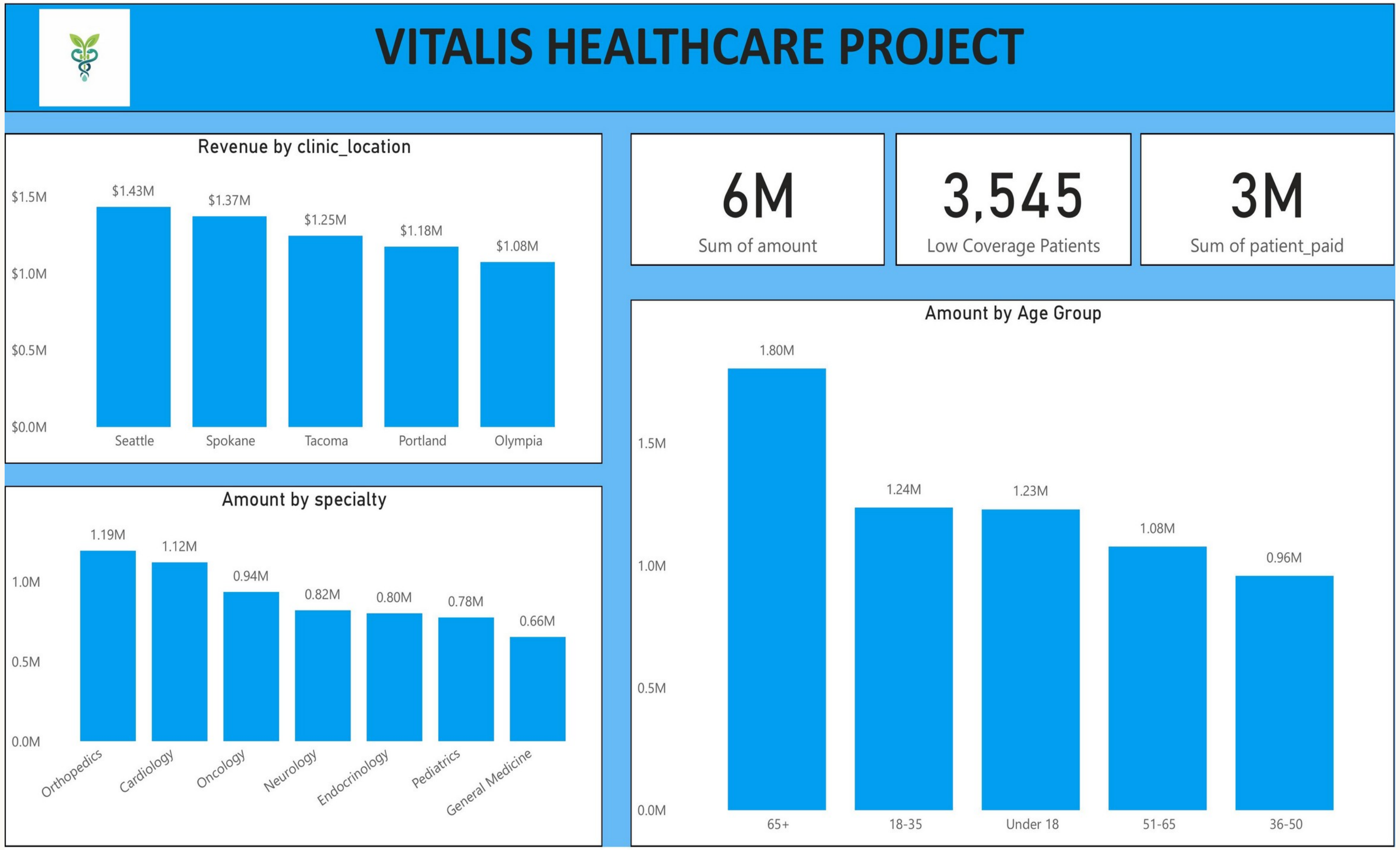
What the patients have



KEY OBSERVATIONS

1. 15,000 diagnoses recorded across the patient population
2. 4,921 patients carry a hypertension diagnosis (I10)
3. 3,533 patients have multiple concurrent diagnoses
4. Diagnosis counts are roughly even across codes — no single condition dominates

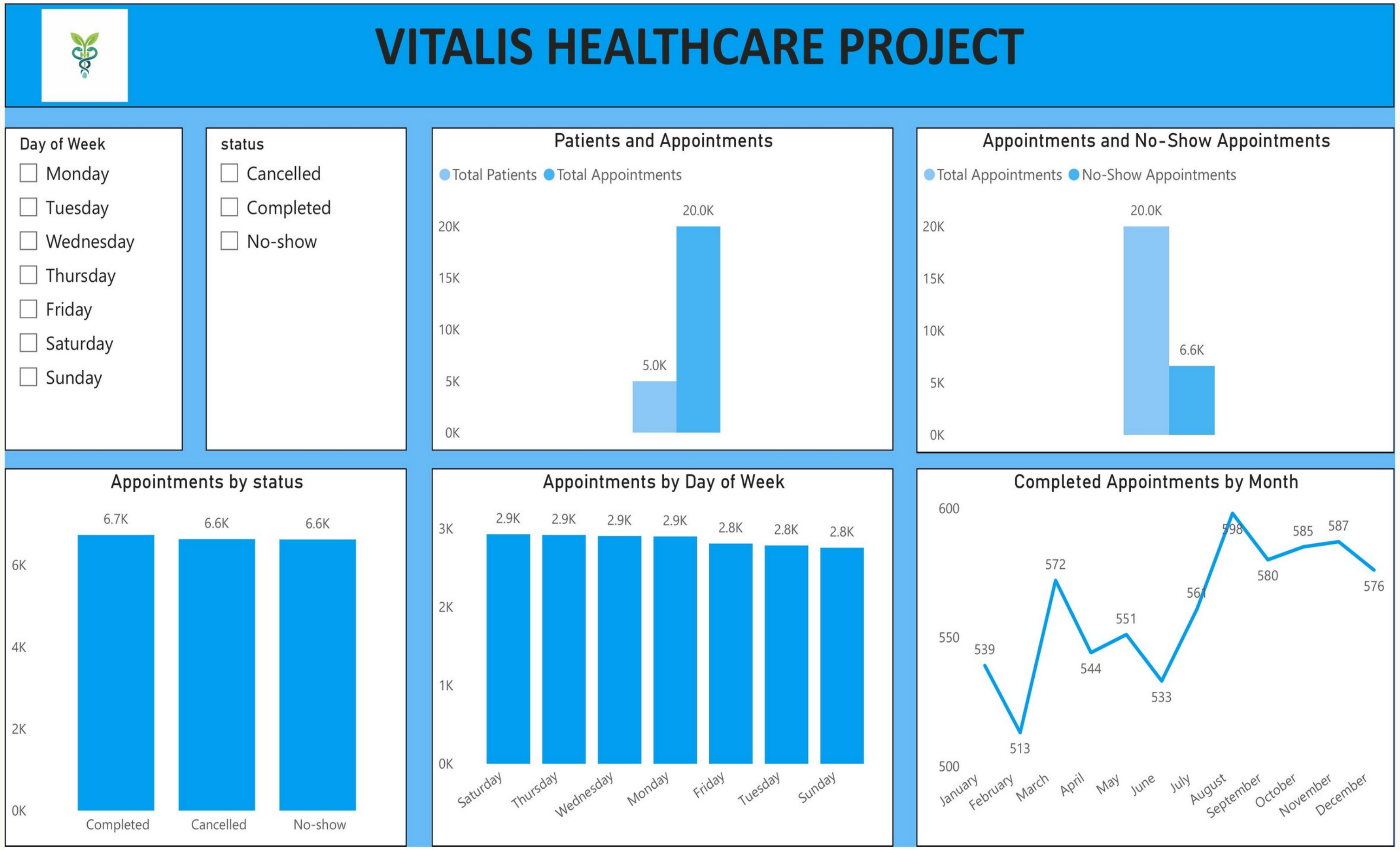
How the money flows



KEY OBSERVATIONS

1. $6M total billed · $3M paid by patients · 3,545 patients with low coverage
2. Age 65+ generates the highest billed amount: $1.80M
3. Orthopedics + Cardiology together drive $2.31M (39% of revenue)
4. Seattle clinic outperforms Olympia by 32% in revenue

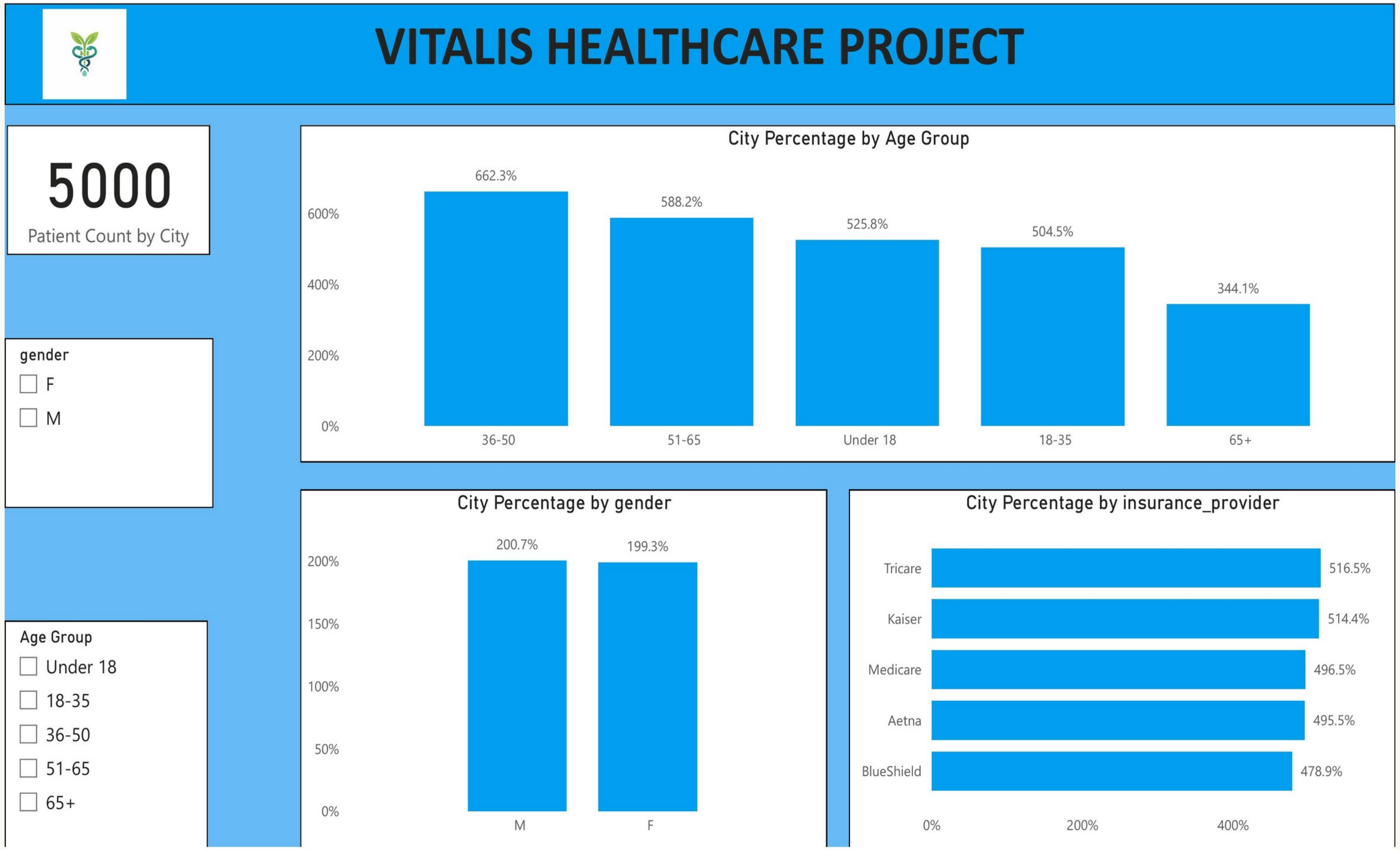
How the practice runs



KEY OBSERVATIONS

1. 20,000 total appointments · 6.6K no-shows (33% — major operational drag)
2. Appointment volume is flat across weekdays (2.8K–2.9K) — no peak day
3. Completed appointments peaked in August at 598
4. Status distribution is nearly even: 6.7K completed / 6.6K cancelled / 6.6K no-show

Population distribution



KEY OBSERVATIONS

1. All 5,000 patients accounted for across cities
2. Age 36-50 is the largest cohort by city share
3. Gender distribution within cities is near 50/50 (M 200.7% vs F 199.3% summed)
4. Insurance providers split evenly — Tricare and Kaiser slightly ahead

What the data tells us

$1.43M

Seattle leads

top-grossing clinic location

33%

No-show rate

~6.6K missed appointments — costly

$1.19M

Orthopedics tops

highest-revenue specialty

4,921

Hypertension patients

98% of the active patient base

$1.80M

Age 65+ revenue

highest-spending age cohort

3,533

Multi-diagnosis pts.

carry more than one chronic condition

What to do about it

Five concrete actions, ranked by expected impact.

01	**Reduce no-show rates**	Implement automated SMS/email reminders and easier rescheduling. Industry data suggests this cuts no-shows by ~30%, recovering significant capacity.
02	**Strengthen chronic disease management**	Develop targeted programs for hypertension and diabetes. 3,533 patients carry multiple diagnoses — coordinated care reduces complications and repeat visits.
03	**Optimise high-revenue specialties**	Invest in Orthopedics and Cardiology (the top two revenue producers): specialist staffing, equipment upgrades, throughput improvements.
04	**Improve lower-revenue locations**	Audit Olympia and Portland (lowest-grossing clinics): access, staffing, specialty mix. Localised improvement plans can close the gap with Seattle.
05	**Enhance preventive care for 65+**	Patients 65+ generate the highest treatment costs ($1.80M). Expand screening and preventive programs to manage long-term cost and improve outcomes.

How the project is built

Three pieces fit together: data layer, analytics layer, presentation layer. Each can be developed and deployed independently.

01 Data layer

SQL

PostgreSQL schema with 6 tables.

17 analytical queries answering business questions about patients, doctors, revenue, diagnoses.

Common pitfalls (integer division, name collisions in GROUP BY, lexicographic comparison) flagged and fixed.

02 Analytics layer

Power BI

6 interactive dashboard pages (Demographics, Specialty, Diagnosis, Financial, Operations, Patient).

23 DAX measures: counts, ratios, period comparisons, classifications.

Sort-by columns for chronologically-ordered axes (Age Group, Day of Week, Month).

03 Presentation layer

Vitalis Web App

React + Vite + Tailwind + Recharts.

Mirror of the Power BI dashboards with cross-filtering, light/dark mode, LocalStorage persistence.

Three pages: Dashboard, Add Record, Settings — all built on the same dataset.

VITALIS

Summary

KEY TAKEAWAY

Healthcare data, transformed into action.

This project demonstrates how patient, financial, and operational data can be transformed into actionable insights using SQL, Power BI, and a modern web stack.

The analysis surfaces strong revenue performance in specific locations and specialties, operational inefficiencies (notably a 33% no-show rate), and the disproportionate impact of chronic disease and aging populations on healthcare costs.

Acting on these findings can improve patient outcomes, operational efficiency, and financial sustainability simultaneously — and the dashboards make those decisions easier to make.